TITLE: COORDINATION AND COMMUNICATION TOOLS AS DETERMINANTS OF PROGRESS IN F/OSS PROJECTS: A LONGITUDINAL STUDY

Summary

Free/Open Source Software (F/OSS) has emerged as an important phenomenon of present times with a growing trend towards considering it as the future development strategy. The Coordination and Communication (CC) Tools used by F/OSS developers form an integral part of the development process itself apart from development technologies being used. The volunteers may join projects at any phase/stage of development and developer pool of projects may change over time. F/OSS volunteers can adopt any role as per their skills.

A *Four-Phase Sampling Process Model* has been developed to select appropriate F/OSS projects for research. Data has been collected for System Software projects and Allied Applications projects (100 projects) from SourceForge Research Data Archive (SRDA), Subversion Logs of projects' code repository and SourceForge.net project pages.

Metrics have been devised to study the intensity of CC Tools being adopted and used by F/OSS project on the whole and their impact on its developmental progress, volunteers' preference for a particular set of CC Tools among the total set being used by a project and the impact of project category and volunteer types on choice of CC Tool have been analyzed using longitudinal panel data models. A web-based tool namely, Coordination and Communication Metrics Reporting Tool (CCMRT) has been developed to compute and report the metrics formulated for the study.

Finally, several problems have been identified which need to be resolved for improving coordination and communication among volunteers as well as better historical trace of data for future research and assessment of F/OSS projects. Corresponding suggestions have been made for improvement.